Sarah Drain

Ph.D. Candidate in Economics, West Virginia University

Urban and Regional Economics | Health Economics | Spatial Econometrics

sarah.drain@mail.wvu.edu | Morgantown, WV | (724) 993-3905

EDUCATION

West Virginia University, Morgantown, WV *2022–Present*

Ph.D. in Economics *Expected May 2027*

Dissertation: *Economic Incentives and Real Activity: Essays on Policy Transmission and Measurement*

Committee: Jane Ruseski (Chair), Alicia Plemmons, Scott Schuh, Kyle Hood

M.A. in Economics *2025*

Duquesne University, Pittsburgh, PA *2021*

B.S. in Economics

RESEARCH FIELDS

Primary: Urban and Regional Economics; Healthcare Regulation

Secondary: Spatial Econometrics; Economic Measurement

JOB MARKET PAPER

The Cost of Care Substitution: Nurse Practitioner Autonomy and Medicare Spending

Examines whether permanent nurse practitioner full practice authority changes Medicare billing participation, claim-recorded NP and physician service allocation, utilization, and standardized spending. Uses staggered state adoption and public Medicare Geographic Variation, Provider, and Provider-Service data alongside AHRF/NPPES records. Distinguishes expanded legal authority from observed billing activity, recognizing that incident-to billing can attribute NP-furnished care to physicians.

SELECTED RESEARCH

Where Does Investment Land? Satellite and Street-Level Evidence on Physical Investment

with Mustahsin Aziz, Kyle K. Hood, Heather Stephens, and David Langerman; Dissertation Chapter 2

Uses satellite and street-level imagery with a spatial border difference-in-differences design to study the location and composition of physical investment following Opportunity Zone designation.

Peer-Reviewed Publication: Alicia Plemmons, Darwyyn Deyo, and **Sarah Drain**. 2025. "The Effect of Certificate-of-Need Laws on Substance Use Disorder Care for Vulnerable Populations." *Southern Economic Journal* 92(1): 111–132. doi:10.1002/soej.12696.

Prescriptive Authority Expansion, Malpractice Payments, and Adverse Actions

with Alicia Plemmons; Working Paper

Evaluates whether state expansions of prescribing authority for psychologists affect malpractice payments and adverse actions using 1990–2024 National Practitioner Data Bank records and staggered difference-in-differences methods.

Chiropractic Scope of Practice: A New Index and Database of Regulatory Stringency

with Alicia Plemmons and others; submitted to the Eastern Economic Journal

Develops a longitudinal state-level measure of chiropractic scope-of-practice restrictions and examines how regulatory stringency varies across states and over time.

Variations in Physical Therapy Autonomy and Provider Density

with Alicia Plemmons and others; submitted to the Eastern Economic Journal

Examines whether differences in physical-therapy practice autonomy are associated with provider density and the geographic distribution of physical therapists.

Chiropractic Scope-of-Practice and Provider Distribution in Pennsylvania

with Alicia Plemmons and others; Knee Regulatory Research Center Policy Brief, 2026

Summarizes how chiropractic scope-of-practice regulation relates to provider distribution in Pennsylvania for a policy audience.

Revolving Credit and the Household Liquidity Channel of Monetary Policy

with Tamás Briglevics, Scott Schuh, and Shaun Gilyard; Dissertation Chapter 3

Studies heterogeneous household cash-demand responses to monetary tightening using linked SCPC–DCPC microdata and geographic deposit-rate variation.

TEACHING EXPERIENCE

West Virginia University

- Instructor of Record, ECON 201: Principles of Microeconomics *Spring 2026*
- Teaching Assistant, BADM 701: Research Methods *Fall 2025*
- Instructor of Record, ECON 202: Principles of Macroeconomics *Summer 2023*
- Teaching Assistant, ECON 202: Principles of Macroeconomics *2022–2023*

RESEARCH & PROFESSIONAL EXPERIENCE

Knee Regulatory Research Center – West Virginia University Morgantown, WV
Graduate Student Research Assistant *2023–Present*

- Construct and validate longitudinal state-level legal databases and multidimensional regulatory indices for licensed health professions, translating statutes and regulations into reproducible measures of scope-of-practice restrictions and policy effective dates.
- Contribute to empirical health-policy research on occupational regulation, provider supply and mobility, patient safety, and access to care, including studies of scope-of-practice and prescriptive-authority reforms.

U.S. Bureau of Economic Analysis – Regional Economic Accounts Suitland, MD
Economist; Regional Income & Value Added *2025–Present*
Economist (Pathways Intern); Research and Analysis Group *2023–2025*

- Contribute to experimental subnational economic measurement research, exploring geospatial, imagery-based, and administrative data as auxiliary indicators of fine-scale economic activity and evaluating their potential for geographic allocation and statistical validation.
- Contributed to research on geographically detailed private fixed investment in equipment, integrating national-account measures with industry-level data and state allocation indicators to evaluate regional investment estimates.
- Evaluated source data and geographic allocation methods underlying official state GDP-by-industry estimates, emphasizing historical comparability, statistical consistency, and reconciliation with national economic accounts.
- Developed reproducible Python and SQL workflows for processing administrative data, validating geographic allocations, detecting anomalies, and evaluating regional economic statistics across production vintages.

RAND Corporation – Arroyo Center Pittsburgh, PA / Washington, DC
Summer Associate, Economics, Sociology, and Statistics Department *2025*

- Conducted federally sponsored health-workforce research on physician recruitment, career incentives, and recruitment barriers in Army Medicine as part of a multidisciplinary research team.
- Developed discounted cash-flow and net-present-value models comparing military and civilian physician career pathways; contributed to stated-preference survey design and analysis; and translated findings into sponsor-facing policy options and research briefings.

RESEARCH SOFTWARE & SPATIAL DATA TOOLS

UrbanEye: Computer Vision for Urban and Spatial Economic Analysis
with Mustahsin-Ul Aziz; Research Software, In Development

Co-developing a Python-based computer-vision platform that uses street-level imagery to construct fine-scale indicators of urban infrastructure and the built environment. The project aims to improve spatial economic measurement by generating geographically detailed data for studying neighborhood conditions, urban development, and place-based policies.

QUANTITATIVE METHODS, DATA, & COMPUTING

Methods: Causal inference; machine learning; spatial analysis; program evaluation; difference-in-differences; staggered-adoption and event-study designs; panel-data methods; applied econometrics; survey and stated-preference analysis; small-area estimation; forecasting; .

Programming: Python, ArcGIS/QGIS, R, Stata, SQL, Git, MATLAB, Julia, Rust, SAS, LaTeX.

Data Engineering & Validation: Data cleaning and harmonization; schema interpretation; record linkage; reconciliation; anomaly detection; provenance tracking; reproducible quality assurance.

RESEARCH GRANTS & FUNDED PROJECTS

- **Case Western Reserve University Nurse Leadership Academy** *2024*
  Nurse Practitioner Research — Sub-award Extension; Research Associate ($60,000)
- **Case Western Reserve University Nurse Leadership Academy** *2023*
  Nurse Practitioner Research — Sub-award; Research Associate ($20,000)

SELECTED FELLOWSHIPS & PROFESSIONAL DEVELOPMENT

- Residency Fellowship for Early-Career Scholars, Institute for Humane Studies *2026*
- DC Urban Economics Day, George Washington University *2025, 2026*
- Using Research to Craft Policy, Institute for Humane Studies *2024*
- Successfully Navigating Your Ph.D., *CSWEP* *2024*
- Women in Economics Symposium, Federal Reserve Bank of Cleveland *2020*

SELECTED CONFERENCES & PROFESSIONAL MEETINGS

- Western Economic Association International Annual Conference *2026*
- American Economic Association Annual Meeting (ASSA) *2026*
- Southern Regional Science Association Conference *2025, 2026*
- Southern Economic Association Annual Meeting *2022–2025*
- North American Regional Science Council Conference *2025*
- NABE Technology Economics Conference *2025*

PROFESSIONAL SERVICE

- Employee Advisory Committee, U.S. Bureau of Economic Analysis *2024–2026*
- Behavioral Economics and Situational Testing Lab, West Virginia University *2022–2024*

REFERENCES

Jane E. Ruseski, Ph.D.
Department Chair
Professor of Economics
West Virginia University
Jane.Ruseski@mail.wvu.edu
Dissertation Chair

Kyle K. Hood, Ph.D.
Chief Economist
Research and Analysis Group
Regional Economic Accounts
U.S. Bureau of Economic Analysis
Kyle.Hood@bea.gov

Alicia Plemmons, Ph.D.
Director
Knee Regulatory Research Center
West Virginia University
Alicia.Plemmons@mail.wvu.edu

Scott D. Schuh, Ph.D.
Associate Professor of Economics
West Virginia University
Scott.Schuh@mail.wvu.edu